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Global Health Interventions: The Military, the Magic Bullet, the Deterministic Model—and Intervention Otherwise

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Abstract

“Intervention” is central to global health, but the significance and effects of how intervention is practiced are often taken for granted. This review takes interventions into health and medicine as subjects for ethnographic inquiry. We highlight three lines of anthropological contributions: studies of global health interventions that serve imperial and military objectives, studies of “magic bullet” interventions arising from laboratory science, and studies of interventions based on deterministic modeling techniques. We then outline examples of “intervention otherwise,” in which people build relations of solidarity and care through global health programming, design interventions to be interactive and adaptable, and use data and modeling to support health justice. Whereas many global health interventions reproduce Western power hierarchies, intervention otherwise draws attention to alternative forms of knowledge, action, and expertise. Our analysis of lively and multivalent practices of intervention has implications for debates about the im/possibility of decolonizing global health.

English: Intervention, the action of intervening; interference, by a country in another's affairs; action, taken to improve a situation, especially a medical disorder;

Spanish: *La intervención* (intervention); *injerencia*, interference or interjection into, e.g., a colloquium or conversation;

French: *L'intervention*, speech, contribution; *l'entremise*, matchmaking; *l'interruption*, break, disrupt, discontinue, stop;

Portuguese: *A intervenção*, interference, mediation, instrumentation; *a interposição*, interposition, interposal; *a intromissão*, intrusion, meddling;

Dutch: *Interventie*, intervention; *tussenkomst*, interference, interposition, agency; *bemiddeling*, mediation, intermediary, procurement;

Somali: *Gargarka caafimaadka*, health intervention; *samaful*, humanitarianism, charity, and care;

Tagalog: *Pamamagitan*, mediation, intercession, intervention; *pakikialam*, caring, meddling, intervening;

Q'eqchí: *Li tz'aq'onk chirix li xkolb'al li kawilal*, to enter, well-being;

K'iche': "It depends"; *Ch'awen to'onem chirij ucb'ajcb'ojil k'aslemal*, to support, healthy life.

1. INTRODUCTION: INTERVENTION AS A SOCIAL PHENOMENON

One popular account of global health describes the field as a "collection of problems" that solidified at the start of the twenty-first century, in which a group of people—mostly white men, mostly from the so-called Global North—adopted an approach to health intervention that supplanted the twentieth-century field known as international health (Farmer 2013; see also Brown et al. 2006). In this account, global health comprises a coalition of philanthropic, governmental, nongovernmental, corporate, clinical, and university experts charged with confronting medical pathologies and threats to biosecurity. Packard (2016) writes in his history of the field that "global health" has been constituted as a series of "interventions into the lives of other peoples."

This history reflects the establishment of global health institutions and the production of global health knowledge and expertise. One study finds that more than 85% of the organizations active in global health or global health policy are headquartered in Europe or North America, and more than 80% of the leaders of global health organizations are nationals of high-income countries, with women holding less than 5% of leadership roles (Davies et al. 2019; Glob. Health 50/50 2020, pp. 17, 26, 30; van Daalen et al. 2020). Racism within global health institutions compounds gender, geographic, and class disparities (Büyüm et al. 2020, Yam et al. 2021). Bhakuni & Abimbola (2021) find that even within the academic field of global health, authorship practices, research partnerships, publishing choices, research questions asked, and methods used, "are peppered with epistemic wrongs that lead to or exacerbate epistemic injustice" (p. e1465). Reflecting on how poorly the United States and United Kingdom handled the outbreak of COVID-19, Dalglish (2020) argues that "the pandemic has given the lie to the notion that [global health] expertise is concentrated in, or at least best channeled by, legacy powers and historically rich states" (p. 1189; see also Manderson et al. 2021).

Anthropologists have long criticized how the field of global health naturalizes a powerful Western vision of what a global health intervention is and on whom it should be operationalized, questioning global health's hegemony and beneficence (Closser et al. 2022, Janes & Corbett 2009, Mendenhall & Closser 2022, Storeng & Mishra 2014, Yates-Doerr & Maes 2019). "Global health" does not just target "Other" people, as Packard (2016) emphasized, but also shapes wealthy and donor countries (Biehl & Petryna 2013, Lock & Nguyen 2018, Nichter 2008). And while leaders may be mostly men from Europe and North America, the care providers, clinicians, aid workers, and advocates implementing global health interventions are more frequently women and people

from racialized minorities and Global South communities (Carruth 2021, Maes 2017, Minn 2022). Global health interventions often present themselves as singular or universal, even while they are, in practice, carried out in different ways to different effects (Redfield 2018).

Our *Annual Review* article thus takes “intervention” squarely as its focus of analysis, following the call of feminist anthropologists to study intervention as a social phenomenon deserving of sustained anthropological inquiry (Benton 2015, p. 10; Ginsburg & Rapp 2013). In doing so, we highlight multiple forms and effects of global health intervention: interventions styled on militaristic aspirations to conquer, control, and secure; magic-bullet interventions that arise from experimental laboratory science; and deterministic global health modeling techniques that undermine local forms of expertise and organizations of knowledge.

Finally, we draw attention to intervention strategies in global health that we are calling “intervention otherwise.” Academics have theorized the otherwise as “an other way of thinking, *un paradigma otro*, the very possibility of talking about ‘worlds and knowledges otherwise’” (Escobar 2007, p. 179; see also Harding 2017, McTighe & Raschig 2019, Meek & Morales Fontanilla 2022). To theorize “otherwise” is to be critical of how white European-American modes of thought and action are consistently recentered. However, the otherwise is defined not only by critique. It also entails a commitment to amplifying practices that are vital and vibrant but which lack the recognition and power of dominant institutions. When it comes to intervention, Mkhwanazi (2016) reminds us that alongside any single story of global health are the “deliberately ignored” stories whose telling might provide inspiration for reimagining the practices of the field of global health (p. 200).

We have started this review with the Oxford English Dictionary’s tautological definition of intervention: “the action of intervening” (<https://www.oed.com/>). We have also translated and back-translated intervention through other languages that we as authors have used in our own work to highlight a range of practices that also, simultaneously, constitute and signify global health intervention. Alongside the ways that intervention figures as a targeted action directed from one site toward another, intervention also includes informal, interdependent, and affective networks and interpersonal relationships. The term’s slipperiness and its multivalent translations through different languages offer methodological and theoretical openings (Law & Mol 2020). In addition to the valuable work of critiquing hierarchically organized, top-down global health interventions, we can also work to amplify how people are using global health to build and remake new kinds of social relations of health and care.

2. INTERVENTION AS MILITARISM, MAGIC BULLETS, AND METRIC MODELS

2.1. Global Health as an Act of War: Imperialism, Military Power, and National Security through Intervention

Colonial medicine responded to the needs of empires, keeping soldiers, administrators, and laborers economically productive across vast geographic expanses (Anderson 2006, Bhakuni & Abimbola 2021, Greene et al. 2013, King 2002). Threats of contagion during imperial conquests, wars, natural resource extraction, and capital accumulation fueled many scientific advancements in public health, immunology, bacteriology, and parasitology, such as the identification of *Anopheles* mosquitoes as the vector for malaria and the discovery and testing of an effective tuberculosis treatment regimen. The interpolation of colonized and enslaved populations into emergent scientific regimes also entailed the pathologization of existing pluralistic and Indigenous systems of healing and medicine and the characterization of “Other” health cultures, beliefs, and practices as unhygienic, uncivilized, and ignorant (Briggs & Mantini-Briggs 2004, Vaughan 1991).

In the twentieth century, following world wars, wars for colonial independence, and the establishment of an international system built on relationships between sovereign states—including first the League of Nations, then UNICEF, the World Health Organization, the World Bank, and numerous international organizations such as Doctors without Borders and Save the Children—global health interventions became central to postwar reconstruction, economic development, and Cold War and postcolonial politics. Even so, most political science and international relations scholars used the word intervention to narrowly index military operations driven largely by national security objectives (Finnemore 2003).

Following the anthrax attacks in 2001 and the worldwide severe acute respiratory syndrome (SARS) epidemic in 2002, it became common to talk about the “global” spread of pathogens and the fact that “disease knows no borders” (de Bengy Puyvallée & Kittelsen 2018; see also Lakoff & Collier 2008). The twenty-first-century field of global health, in contrast to international health in the twentieth century, was characterized as *sans frontières* (without borders), even as interventions continued to be designed and funded to increase national security and often involved engagement with or funding from military, defense, and police forces (Kline 2019). The material infrastructures and human architectures of global health—its labor hierarchies, field operations, chains of command, cold chains, logistics, donor–recipient relations, financing models, and so on—still echo and employ imperial and military hierarchies and activities. The militarized origins of global health also remain evident in global health vernacular: We “fight” the potential “invasion” of contagious pathogens; we “fortify” nutritional supplements; and we “deploy” workers to the “front lines” of epidemics. Illnesses, much like enemies, are “targeted” for control, elimination, and eradication.

As such, US President Barack Obama dispatched the 101st Airborne Division to “fight” Ebola, and Sierra Leone’s President Ernest Bai Koroma declared a “military approach” to Ebola in 2014, appointing his former defense minister, a retired army major, as head of the National Ebola Response Centre and urging citizens to use force against those who resisted public health directives (de Waal 2014). “Enemy attack” was also a key framing during the response to COVID-19 (Lutz & Crawford 2020, Yates-Doerr 2020). “The beguiling metaphor of sending soldiers to fight pathogens still wins out,” de Waal (2014) argues, at least for garnering political support and funding, “fueled by our deepest fears of disease, and by our uncritical acclaim for soldiery.” The US Department of State (2022) still explicitly links health interventions and national security: “Halting and treating diseases at their points of origin is one of the best and most economical ways of saving lives and protecting Americans. The U.S. National Security Strategy and U.S. National Biodefense Strategy prioritize U.S. efforts to build global health security capacity.”

Global health interventions have not only long targeted contagions at their point of origin, but also targeted people crossing borders, treating them as vectors of disease and, thus, as a site for fear, surveillance, and control. Immigrants crossing land and sea borders into countries in North America and Europe in the nineteenth and twentieth centuries, for example, were subject to mandatory chemical sanitation or sprays, forced immunizations, and invasive medical examinations (Bashford 2006). Migrants have long been subjected to testing and detention for the most feared epidemics of the day: tuberculosis in the nineteenth and twentieth centuries, HIV in the late twentieth century, SARS and Ebola at the start of the twenty-first century, and most recently COVID-19 (Keshavjee & Farmer 2012, Martinez 2021, Mason 2012, Storer et al. 2022, Wynn 2021).

As the field of global health uses the militaristic vernacular, goals, and organizational forms from its predecessors, global health has also been part of myriad diplomatic negotiations, foreign policy strategies, and social movements (Al-Mohammad 2019, Hamdy & Bayoumi 2016, Mazzarino et al. 2019, Pfeiffer & Chapman 2010, Pike et al. 2010, Rylko-Bauer et al. 2009). Clinicians with Médecins Sans Frontières, for example, practice witnessing, or “*témoignage*,” to advocate

and speak out publicly for patients' rights and political actors' responsibility for providing health care, especially in wartime and in the face of human rights abuses (Redfield 2013). During the COVID-19 pandemic, Lasco et al. (2022) found that clinicians acted on the "political frontlines" to provide clinical care, but also acted as advocates for better health policies, more resources, and better political leadership.

Accordingly, global health interventions and health providers are frequent targets of military and defense operations (Fouad et al. 2017, Haque et al. 2022, Omidian 2016, Omidian & Panter-Brick 2015, Redfield 2016a). In 2011, the United States Central Intelligence Agency used a door-to-door vaccine intervention to verify the whereabouts of Osama bin Laden, leading to his killing—and, in the years afterward, exacerbating distrust in vaccination campaigns and violence against health care workers across the region (Closser & Coburn 2019). The Government of Ethiopia, from November 2020 until the writing of this article, has blocked shipments of medical supplies from entering the contested Tigray Region, and multiple parties to the conflict there have targeted clinics and aid workers delivering health services (Gesese et al. 2021). Despite violating the Geneva Conventions and other international laws, attacks on and manipulations of health interventions are essentially force multipliers: They can foster widespread fear and distrust of health care, depriving people of life-saving services, and leading to more harm than direct violence alone. From conquests and colonialism of centuries past to the present day, health and military interventions have been consistently intertwined and interdependent.

2.2. Global Health as Reductive, Linear, and Unidirectional: The Magic Bullet Intervention

The wooden ships of imperial capitalism were early laboratories of experimental science. Isolation for months at sea and rigid sailing hierarchies allowed ship doctors to examine cause-and-effect variables on the vulnerable bodies of sailors or enslaved people. One popular story credits the Scottish sailor-surgeon James Lind as the progenitor of experimental medical intervention. Supposedly, as his crew grew sick, he provided a citrus elixir to only half, who recovered, paving the way to the eventual discovery of scurvy along with its cure. However, Milne (2012) points out that many sailors already knew to eat citrus at sea, cause-and-effect trials on human bodies had been going on elsewhere for centuries, and Lind never actually recommended citrus to cure scurvy. However, Lind's maritime laboratory is still held up as the origin of experimental health interventions, which speaks to the powerful role of myth in claims about the success of elixirs—or "magic bullet" interventions.

Four centuries later, health professionals continue to throw their energies into magic bullet treatments (Cueto 2013, p. 30), and anthropologists continue to point to the fallacy and breakdown of this approach. For example, Kelly & Lezaun (2021) document the waning efficacy of domestic antimalarial interventions promoted as a cure for malaria, such as chemical sprays for interior surfaces or insecticide-treated bed nets. Facing increasing malaria, health officials in Tanzania have become excited by the promissory technologies of the "vectorsphere," which relies on fabric ribbons covered with a chemical repellent. Yet the vectorsphere's own inventors have concerns about how the ribbons do little to address the underlying infrastructures of poverty, worrying that "by virtue of their very cheapness and ease of application, they might in fact help perpetuate chronic forms of destitution" (Kelly & Lezaun 2021, p. 570).

Kelly & Lezaun's interlocutors are in good company with global health anthropologists, who highlight the limitations of seeking magic bullet cures to complex social and health problems. Laboratory logics established the figure of the universal patient, on whom health interventions

designed anywhere can effectively be imposed. This figure relies on the assumption that medicines and interventions are universally effective. Yet time and again global health practitioners confront cultural heterogeneity when they bring laboratory logics into everyday life. The release of genetically engineered mosquitoes in Burkina Faso, for instance, failed to reduce malaria rates but instead undermined trust in public health (Beisel & Kuumuori Ganle 2019). Liberian refugees experiencing acute food insecurity and high rates of malnutrition rejected the United Nations World Food Program's food aid because the taste of the food mattered (Trapp 2016). Polio eradication programs in Northern Nigeria ignored a long history of imperialism and distrust between people and governments, thereby failing to address parents' primary concerns about their children's health needs (Renne 2010). Cash transfer interventions in rural Peru left their beneficiaries ever more excluded because of the social regulations that surrounded the aid (Cookson 2018). Again and again, all over the world, promises of universally applicable technological interventions go unkept.

In a powerful critique of elixir-based interventions, Venkat (2021) writes about death from tuberculosis, a disease whose status in global health discourse as curable has become increasingly suspect in the face of antibiotic resistance. He shows how fantasies of eradication run into the contradictions of etiology and effect; the simplicity of cure as imagined in the pharmaceutical laboratory is unable to contend with the complex realities of sickness in everyday life. Venkat cites Fanon to suggest that "cure" is often impossible without a transformation of the social order (Fanon cited in Venkat 2021, p. 11). As Biehl & Petryna (2013) write, global health interventions tend to "leave people behind, not only by limiting access to the services provided but also by producing a parallel system of care and governance that undermines other avenues for care that might take into account broader systemic factors" (p. 135). Lasco & Yu (2022) add that this elixir-based paradigm lends itself to "pharmaceutical messianism," or the invocation by political actors of various substances as simplistic solutions to cure and prevent epidemics. For instance, during the COVID-19 pandemic, politicians touted medicines such as Ivermectin, hydroxychloroquine, and monoclonal antibodies, borrowing from the "epistemological authority" of biomedicine (Lasco & Yu 2022, p. 3; see also Moyer & Nguyen 2016).

Even as they fail to achieve promised results, laboratory logics of intervention that offer simple cures as the answer to complex social problems do not just endure, but travel. Petryna (2009) describes the "offshoring" of randomized controlled trials as part of a global search for human subjects. She notes that pharmaceutical research and development soared from \$1.1 billion in 1975 to \$44.5 billion 3 decades later (p. 12). The global proliferation of pharmaceuticals has made the uncompromised, treatment-naïve bodies necessary for clinical research harder to find, pushing the commercial medical industry to deeply integrate with local health systems and drug markets, expanding the frontiers of magic bullet interventions (Petryna 2009, p. 11; Shah 2006). Valdez's (2021) research on clinical trials raises profound questions about why randomized trials maintain their prominence when they so often fail to improve people's lives. She points to the power of racial capitalism and *cis*-heteronormative-white supremacy to explain why experts push drug trials instead of structural changes such as expanded pregnancy leave, sick leave, and fair wages. Drawing from Benjamin's (2019) analysis of technoscience, we might see the ongoing "failure" of interventions to achieve better health not as a bug but as a feature of global health, which depends on "discriminatory design" (p. 5).

2.3. Global Health as Universal, Scalable, and Deterministic: The Metric Model Intervention

The interest in laboratory health sciences that emerged in the late nineteenth century grew in power and prominence in the twentieth century with the invention of the randomized controlled

trial (RCT). The premise of the RCT is that there is a universalism to human bodies and societies such that a prevention or treatment strategy tested in one place could work anywhere in the world. Health experts treat the RCT as the gold standard of global health: an objective science that would produce universal and scalable interventions, such as antimalarials, vaccines, and antibiotics. Adams (2016) points out that “inventory” and “intervention” share a similar root in her observation that many global health interventions have the goal of expanding markets for corporate products and capital (p. 11). She links the expansion of capital to the growing field of evidence-based medicine that prioritizes econometric statistics and other forms of health accounting reliant on big data. These new evidence technologies have changed the scope and meaning of doing health work, altering how time is spent everywhere from the boardroom to the rural clinic (Adams et al. 2016). Tichenor explains that nation-states have long relied on numerical surveillance to control their population; however, as global health has expanded beyond the nation-state in the 21st century, quantitative reasoning has also extended its reach, with metrics increasingly used to define institutional efficacy, individual happiness, and health (2020). Statistical data analysis has become vital to intervention economies and “appearances of global health prowess and accomplishment” (Erikson 2019a, p. 508; Carruth 2018).

Rottenburg & Merry (2015) show that the rising dependence on global indicators privileges quantitative knowledge when it comes to how both state and nonstate actors distribute attention, make decisions, and allocate scarce resources (p. 3; see also Erikson 2016, Shore & Wright 2015, Stein 2018). They point out how a system of numeric analysis that promotes itself as accountable and democratic, in fact, encodes particular cultural understandings and political and economic interests that reproduce inequalities. Consider the damaging effects of data collection technologies imposed by United Nations maternal health interventions as documented by Suh (2021) and Strong (2020). Senegalese health experts were so focused on numerically demonstrating the efficacy of their model for postabortion care that they ignored the unsafe abortion care that many women received (Suh 2021). In Tanzania, there was so much pressure to document good care practices that women were left without interpersonal support (Strong 2020). In both cases, the accountability systems encouraged by the global health community contributed to the maternal mortality they claimed to prevent.

In 2022, the American Anthropological Association (AAA) invited Christopher Murray, the longtime director of the Institute of Health Metrics and Evaluation (IHME)—an influential global health center closely connected to the University of Washington and the Bill and Melinda Gates Foundation—to be a keynote speaker at its annual meetings. The IHME received considerable media acclaim during the COVID-19 pandemic for its statistical models, although their projections were eventually shown to be simplistic, misguided, and lacking context (Jewell et al. 2020). A petition signed by more than 80 anthropologists resolutely objected to Murray’s invitation, noting “[t]he decision to give Murray a platform at the AAA meeting is a tacit endorsement of methodologies which are neither consistent with anthropological critiques of power nor complementary to anthropological approaches to health and medicine” (Manderson et al. 2022). The AAA president responded by rescinding Murray’s invitation, recognizing the need to critically interrogate the underlying structures of privilege that encourage “the datafication of health” (Ruckenstein & Schüll 2017). The revocation speaks to a call to pay more attention to how claims of culturally normative moral authority within global health become reproduced (Beaulieu 2022, Brada 2016, Cuj et al. 2021, Gore & Parker 2019).

3. INTERVENTION OTHERWISE

Anthropologists have illustrated, through numerous ethnographic accounts, how global health experts model interventions on visions of military battlefields, magic bullets, and deterministic

statistical models. These interventions frequently take aim at distant, “exotic” targets, seeking to maximize efficiency, securitize populations, and control and communicate outcomes through various algorithms and metrics. Presuming a universality to the human experience of health and illness, and holding white-*cis*-heteronormative values to be standardizable, experts design interventions to be replicated and scaled up from site to site, treating health as something that can be quantified and controlled, even at a distance and despite the obvious diversity of contexts. Technologies and techniques deployed from powerful centers into distant peripheries accordingly remain unchanged.

In addition to critiquing the shortcomings of interventions premised upon the battlefield, magic bullet, metrics, and models, anthropologists have also helped to illuminate alternative global health epistemologies and ways of envisioning and enacting global health interventions. At the beginning of this article, we translated the term intervention out of and back into the English language to show the very concept of intervention to be lively, multivalent, and unstable as it moves. The Spanish language, for example, helps us to conceptualize an intervention as an interjection into a conversation: not all-knowing officials acting their will upon others, but a provocation, a questioning, or pause. French offers the idea of a contribution, a match-making exercise, or a discontinuation or breakage. Intervention in Portuguese can connote mediation or interposition, along with the more negative connotation of intrusion or meddling. As we move outside of colonial languages, other possibilities also unfold. Tagalog gives us *pakkialam*, which can mean both caring or interfering and is rooted in “*alam*” or knowing. Somalis’ concept of “*samafal*” points toward humanitarianism, care, and solidarity. And when discussing the translation of intervention with Maya-language speakers, they offered the feedback that the proper translation for the term “depends” on the interaction.

3.1. Interference

Haraway (1999) uses the image of diffracting rays of light—where interferences at one point in the movement of light will change its future pathway—to draw attention to how traveling entities do not straightforwardly replicate themselves. Thinking along with interference, we might note how frequently people interfere into global health experts’ visions, remaking interventions on the ground. For example, McKay (2018) describes how care workers in Mozambique creatively redistribute food baskets to people in need. As she illustrates, global health interventions comprise not only high-tech therapeutic materials, but “relations among family, friends, colleagues, and neighbors” (p. 3). Additionally, midwives in Mexico approached global health as a relational practice, one they could strategically harness to strengthen their professional standing and improve care (Dixon 2020). Dixon (2020) writes, “As midwives work with and against the infrastructures, ideas, technologies, attitudes, and assumptions of global health policies and interventions, they move things forward, ever pushing against the grindingly slow gears of bureaucracy and oppression. They do global health, but as they do it, they reshape it and make it their own” (p. 7).

Hardon & Hymans (2016), writing from Amsterdam and the Philippines, illustrate how many global health practices of harm reduction are not driven from top-down directives but, rather, taken up “from below.” They frame these care practices as global because they are spread through transnational networks of communication and support, not because they are imposed from large global institutions. We might similarly see the distribution of misoprostol and mifepristone from activists in Mexico and organizations such as “Women on the Web” to the United States following the repeal of *Roe v. Wade* as a global health intervention (van de Wiel 2022). Here, well-networked abortion defenders saw a critical health need that had emerged in the United States and realized they had the capacity to provide care, thereby interfering both with the restrictive laws in the

United States and with conventional hierarchies of intervention in global health. Finally, Carruth (2021) describes the activities of Somali health and humanitarian aid workers who appropriate humanitarian relief interventions to render them “samafal”: a notion of care and crisis response grounded in familiar enactments of kinship, religious practice, political solidarity, and collective liberation from imperialism and marginalization at the hands of the Ethiopian state. Whereas many global health experts emphasize an individualized cost–benefit equation to COVID-19 prevention measures such as mask wearing, vaccinating, or social distancing (Blume 2020, Mendenhall 2022, Sinha 2020), enactments of samafal transform exogenous global health interventions into locally meaningful forms of power and care.

3.2. Interessement

Akrich et al. (2002) offer the idea of “interessement” in French to highlight how global interventions should be understood as iterative and adaptive. They describe how photovoltaic kits were developed and distributed throughout Africa under the assumption that children would need light for evening schoolwork. French engineers concocted a simple mechanism, prototyped it in a laboratory, and then scaled it up for distribution, only to find that people ignored their kits. “Science discovers, industry applies, and man follows” (Akrich et al. 2002, p. 202) may have been the hubris guiding the engineers’ interventions, but the successful adoption of health technologies requires the interessement work of negotiating and translating between these technologies and people’s sociomaterial environments. De Laet & Mol (2000) similarly take up the case of a bush pump in Zimbabwe, which they call a “fluid” technology, in the sense that flexibility was part of its design. In contrast to the photovoltaic kit, the bush pump depended on community involvement and environmental variation: people and bacterial fluctuations were brought in as part of its “nuts and bolts” mechanics rather than treated as external to them. Reis-Castro (2021) likewise describes antimalaria efforts that rely on a carefully coordinated management of multispecies relations, in which the transgenic *A. aegypti* mosquito is “simultaneously a friend and an enemy, cared for and killed” (p. 323). The adaptive capacity of global health technologies is not something to be straightforwardly celebrated. As Grant (2022), Beisel & Schneider (2012), and Redfield (2016b) note, humanitarian organizations can quickly capitalize on flexibility to further their own interests. But the observation that global health’s interventions do not just work but instead are negotiated and translated remains a key anthropological contribution to global health.

3.3. Matchmaking

Intervention can back-translate to matchmaking, highlighting a pragmatic realignment of need and resources that can be both serious and playful. For example, anthropological studies of global health’s modeling practices show how even standardizable, scalable interventions offered by health experts are also taken up as sources for on-the-ground improvisation, extrapolation, and creative manipulation. Biruk (2018) offers the metaphor of “cooking data” to illustrate how even when—or perhaps, especially with—working with cold, objective, quantitative data sets, people frequently change or differentially interpret and use data to accommodate their needs. Biruk’s ethnography contributes to an emerging movement to merge critique with the observation that justice, too, can go viral (cf. Benjamin 2022). Big data has been widely viewed (and feared) as an instrument of colonial intervention, security, and state control, but, in practice, it is also being envisioned in ways that challenge conventional political configurations of global health (Hall-Clifford et al. 2022). Pelizza (2020) thus shows how data are used by migrants to achieve “visibility” and “de facto inclusion.” Kaul & Yates-Doerr (2023), writing from Bhutan and Guatemala, note that health care professionals are modeling growth by emphasizing a child’s interdependence with its surroundings—a

departure from growth models privileged by Euro-American health care professionals that emphasize individualization and hierarchy. Prussing (2023) highlights a field of postpositive quantitative research informed by Indigenous and proequity epistemologies of data. She draws on research with epidemiologists working outside Western centers to show how communities are generating and promoting accessible data that have benefits for health promotion and Indigenous data sovereignty. “Closer attention to such work helps to enhance ethnographic inquiry into how quantification, positivism, and privilege relate and constructively extends associated support for epistemic pluralism,” Prussing (2023, p. 37) writes. As García-Meza & Yates-Doerr (2020) illustrate, based on research with health workers who are invested in assessing and monitoring the weight of babies in Guatemala as part of a nutrition intervention, even metrics are not fixed but have “social life,” with people putting them to use in locally meaningful ways.

3.4. Interposition

Many global health interventions rely on bringing “global” expertise to “local” practice, but a focus on interpositionality illuminates the complex positionalities involved in interventions, where the expertise of the global health worker, as well as that of the anthropologist, is often turned inside out. Neely & Nading (2017) ask anthropologists to explicitly consider how their own university and institutional affiliations affect the knowledge they produce and disseminate about global health. Do anthropologists serve in the role of observer or witness? In what ways do our objectives align with objectives held by other global health practitioners or the subjects of these interventions? The “lone fieldworker” and “single author” style of ethnography is increasingly critiqued as misrepresentative of anthropological methods, as we rely mostly on collaborative, socially embedded forms of data collection and interpretation (Kotni et al. 2020). Today many anthropologists are instead experimenting with interactive collaboratives—or co-laboratives, as Yates-Doerr (2019), following de la Cadena, has described the labor that goes into bringing together different kinds of expertise in global health. Saldaña, for example, worked with a group of global health scientists and activists to protest a stigmatizing and fat-shaming comic launched by the López Obrador government in Mexico. They collectively published a condemnation of the initiative in a high-profile newspaper (*Animal Politico* 2020). In an interview, Saldaña described the process as challenging, given that the collaborators had distinct disciplinary differences to overcome, but also as an “amazing exercise” in collectively fighting state-sanctioned fat phobia (Yates-Doerr et al. 2022). Benezra is part of a transdisciplinary group of anthropologists and microbiologists/ecologists who share a commitment to developing antiracist scholarship about the human microbiome (Benezra 2023, De Wolfe et al. 2021). Brunson (2022) collaborates with the CommuniVax Coalition to bring medical anthropological insights about racism and inequity into COVID-era health policy, giving anthropologists a “seat at the decision-making table.” Still others have raised questions about the limitations of working within conventional systems of science and governance to achieve community health. Carney et al. (2022) describe a coalition of academics and activists working in Tucson, Arizona, during the COVID-19 pandemic who are calling on the need to address anti-Blackness embedded within existing health systems. Activist Karen Washington has argued that when it comes to improving community health it is not a “seat at the table” that is needed but an abolitionist politics that upholds forms of nourishment not dictated by settler colonialism and slavery (Washington cited in Carney 2021).

3.5. Disruption

Many global health interventions are understood as fulfilling Western desires and politics, but a possible translation of intervention as disruption instead highlights an explicit refusal of the

Western status quo. Writing from Zimbabwe, Gumbonzvanda et al. (2021) describe a decolonial practice of Nhangas based on intergenerational and relational expertise—not conventional global health expertise—that women use to collectively organize and benefit both their communities and global health. Abimbola (2022), the editor-in-chief of the influential journal *BMJ Global Health*, has pushed for changing the ways that global health is legitimized, noting that “[t]here’s nothing normal about doing a study in Nigeria and wanting to publish it in London.” Anthropologists have also called for transforming how knowledge of “global health” is legitimized in anthropology. Mkhwanazi (2023) shows how south-south networks of anthropological expertise on health have emerged among Argentina, Chile, China, Eswatini (formerly Swaziland), Kenya, Lesotho, Mexico, Nigeria, South Africa, Sudan, and Zimbabwe but that this knowledge is often not reproduced in a citational form currently recognized by the Anglo publishing world and a white, Anglophone academy. While the genre of the *Annual Review* tends to highlight written texts, much anthropological work on global health interventions is difficult to cite: policy advocacy, community activism, podcasts, expert witnessing, clinical care, participatory artwork, and more. When we begin to open up the possibilities for intervention, it is not only the field of global health that is expanded. The perimeters of anthropology, who counts as an anthropologist, and how we carry out our interventions change as well.

4. CONCLUSION

Still pretty sure you can’t decolonize global health and still have global health at the end.

—Adia Benton, October 3, 2021

Anthropologists have been highly critical of the militaristic, reductive, linear, universalizing, and deterministic models adopted by many global health interventions, which routinely impose Western strategies and epistemologies that can destabilize or sideline the communities they claim to help. At the same time, anthropological analyses of intervention-in-practice illuminate how global health can be—and is already—imagined and constituted in ways that run counter to these historically dominant practices. In reviewing anthropological scholarship on global health interventions, we clearly see the colonial domination of outsiders who intervene upon (people treated as) “Others.” Yet we also see the interventions of local organizations and experts who spend their time caring for chronic and acute health problems, who are also centrally involved in producing, critiquing, and transforming the terrain of global health.

The idea of a global health intervention in the 1990s typically evoked a tangible object such as a vaccine, medication, or supplement that moved from a developed center into the developing periphery. Anthropologists have offered numerous critiques of this reductionist formulation of global health intervention, arguing for attention to diversity, complexity, and power (White et al. 2002, Whiteford & Manderson 2000). Today, however, many powerful global health actors are well aware of diversity and complexity, and many interventions are anything but simplistic in their design (Erikson 2019b, Hanson 2022, Herrick et al. 2022, Müller 2020, Ulijaszek 2015). Contemporary global health interventions often entail complex and mobile technologies such as market manipulation devices, infodemic management strategies, big-data algorithmic accounting systems, high-tech communication, sophisticated biomedical screening technologies, and transnational and transregional collaborations and movements. Attention to what we are calling intervention otherwise—that is, alternative modalities and relations through which global health is practiced—will not only help to amplify grassroots, non-Western, anti-imperial, and anticapitalist work happening in global health, but potentially also bring attention to changing techniques and technologies of power.

Taking intervention squarely as the focus of analysis gives us insight into how military and colonial practices frequently cause harm not because they are reductionist but because they are designed to ensure the privilege and power of the elite actors behind the curtain of global health. In other words, it is not so much the simplicity of interventions that anthropologists can take to task, but the reproduction of heteropatriarchal and white supremacist networks of power (Chaudhuri et al. 2021, Valdez & Deomampo 2019). For example, many governments in Latin America have developed policies to create interventions inclusive of Indigenous communities, such as the adaptation of traditional medicine, linguistic diversity, and intercultural sensitivity in the training of health providers (Gallegos 2011, Hernández 2022, MSD 2012). Yet even while these interventions seem to humanize health services and care, in practice they frequently do little to dismantle systemic racism, instead pushing assimilation and reducing opportunities for intercultural dialogue (Campos et al. 2017).

We conclude with Benton's (2021) provocation that global health cannot be decolonized and "still have global health at the end." Increasing numbers of professionals and students in the field of global health actively promote decolonization, or the handover of power and resources from institutions in North America and Europe to the people, organizations, and governments located in places where interventions occur (Abimbola et al. 2021, Chan et al. 2020, Peace Direct et al. 2021). Can we consider these acts global health interventions without the recognition or buy-in of those in power? To what extent might this revolution produce global health interventions now unrecognizable and untethered to the histories of war and imperialism that we have described? We take heed of Maya-language speakers' insistence on the importance of context, countering the impulse in global health to find a global answer to these questions. Instead, our review of literature on global health intervention finds that anthropologists are well-positioned to take them up through the situated and relation-centered research our methods enable. The task before us is to expand our own frames of analysis and action, making room for anthropological interventions into global health that are critical, constructive, caring, co-laborative, and other modes of engagement that we have yet to name.

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